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LABORATORY-SCALE PLANETARY SURFACE MAPPING USING USRP-BASED SYNTHETIC APERTURE RADAR

GUIDED BY:

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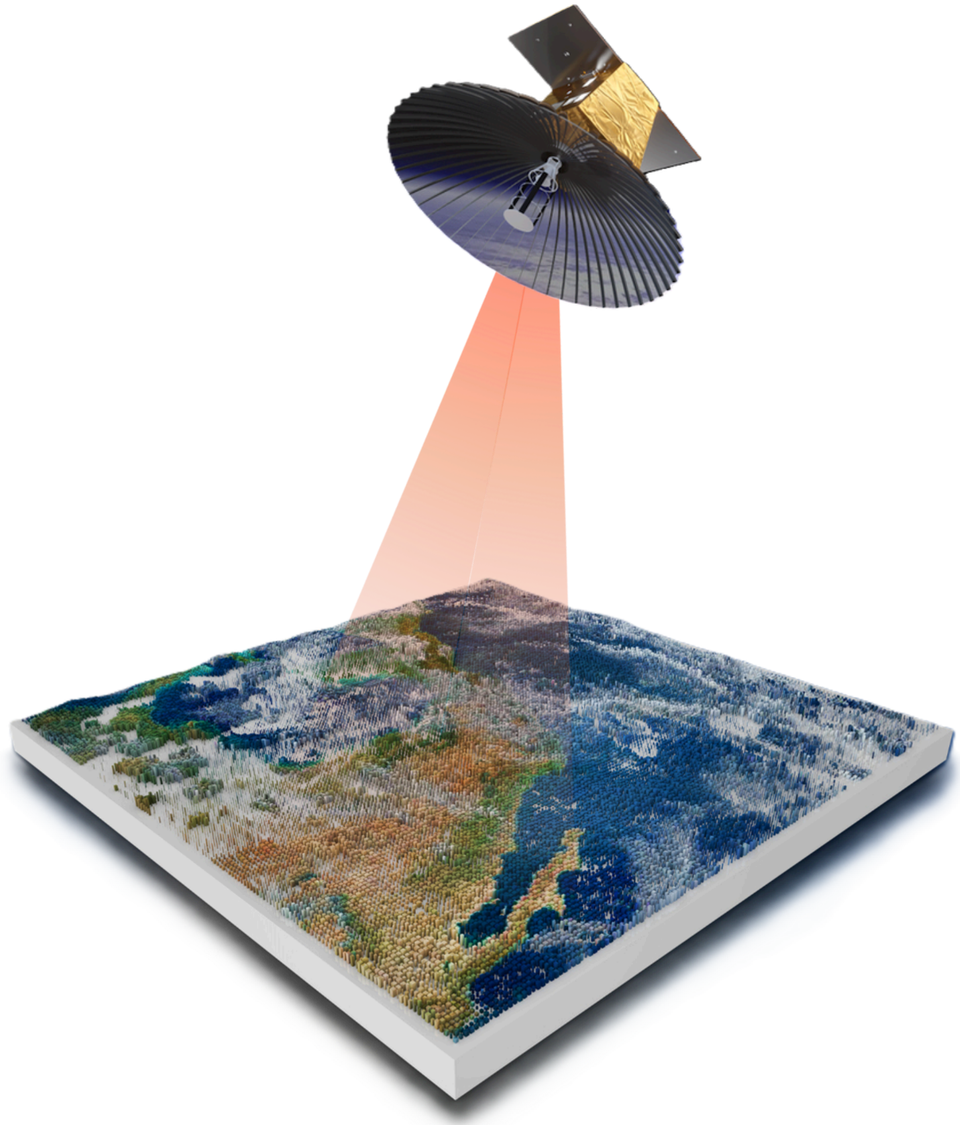
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INTRODUCTION

- **Planetary surface mapping** is essential for understanding the geology, morphology, and physical properties of planets and moons.

- Planetary radar mapping serves as a critical bridge between standard photography and physical landers. Because **RADAR is an active sensor** meaning it provides its own "light" in the form of radio waves it can reveal data that **optical cameras simply cannot see.**



WHY RADAR ?

- Radar-based remote sensing is widely used in planetary exploration because it **can operate in darkness and penetrate clouds, dust, and surface layers** where optical systems fail.

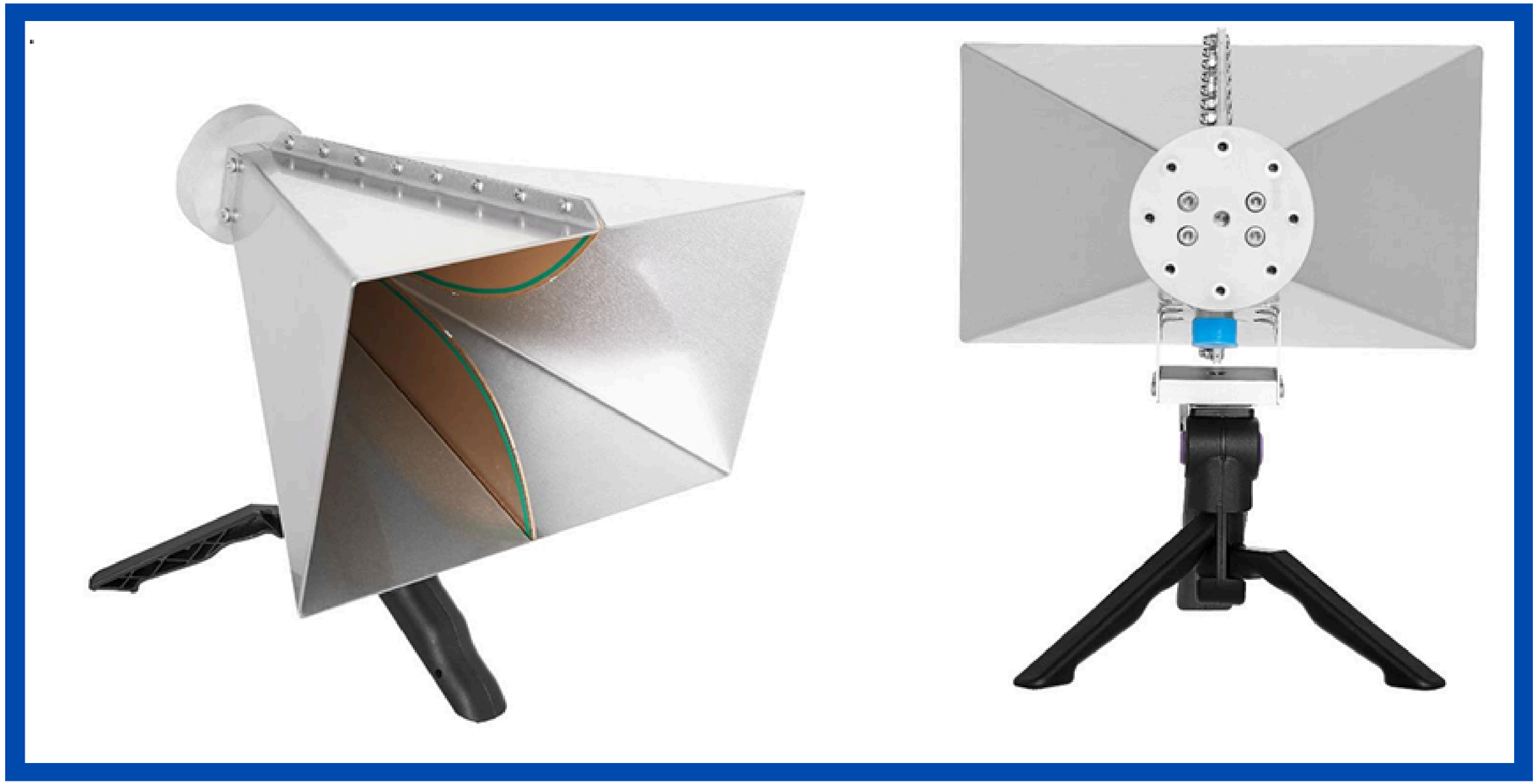
- Microwave radar systems transmit electromagnetic waves toward a surface and analyze the reflected echoes to **extract information such as range, reflectivity, and surface roughness.**

- **Synthetic Aperture Radar (SAR)** improves spatial resolution by coherently combining radar measurements collected from multiple antenna positions.

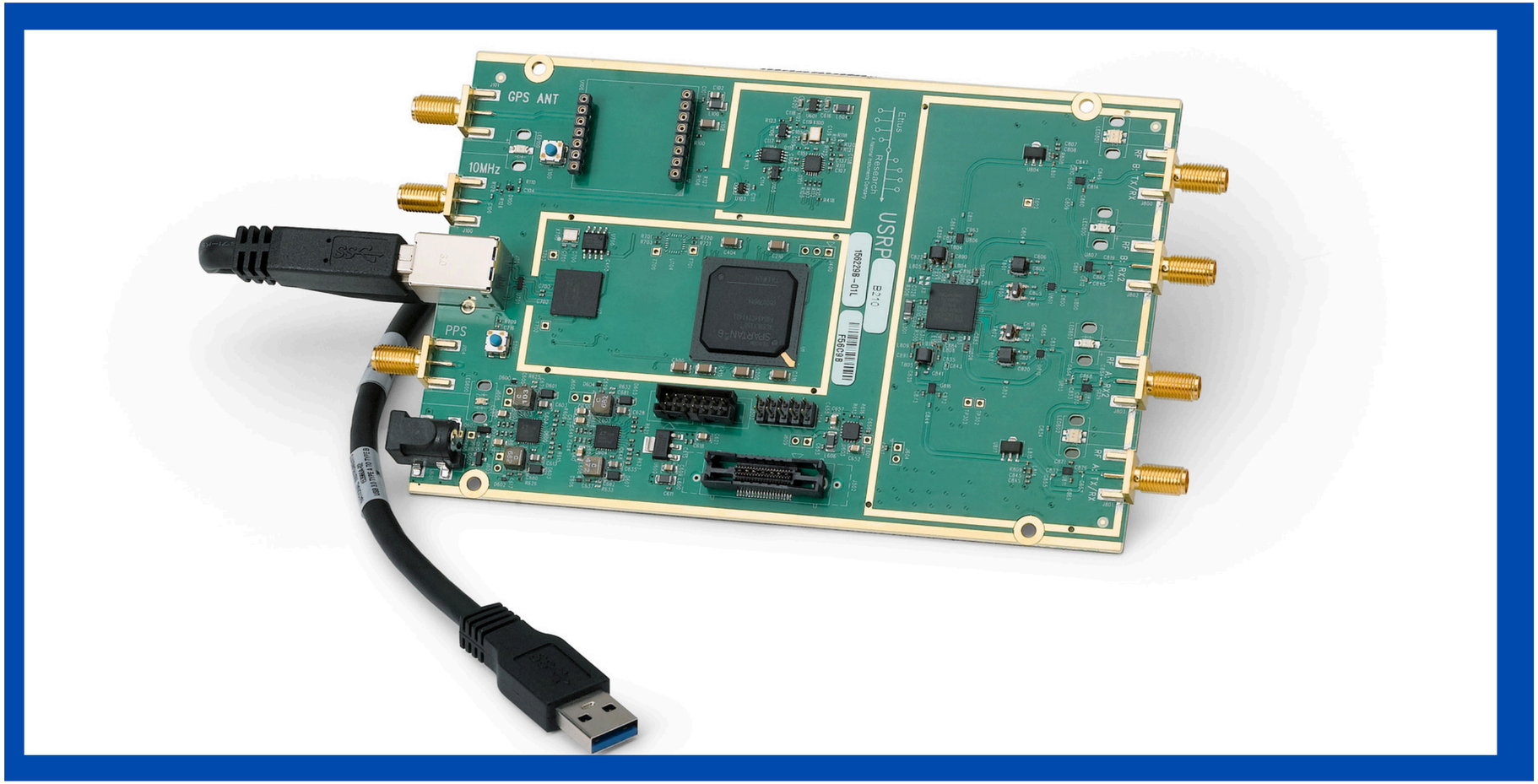
PROBLEM STATEMENT

Conventional planetary surface mapping techniques **primarily rely on optical and imaging sensors**, which are highly dependent on sunlight and are **significantly affected by atmospheric conditions**. These methods provide limited information about surface roughness and material properties and are ineffective for subsurface or all-weather observation. RADAR based techniques, particularly **Synthetic Aperture RADAR (SAR)**, overcome these limitations by enabling high-resolution surface characterization **independent of illumination and environmental conditions**. However, existing RADAR and SAR systems used in **planetary missions rely on highly specialized and costly instrumentation**, which limits their practical implementation at smaller scales. This creates the need for a **low-cost, flexible platform such as the Universal Software Radio Peripheral (USRP)** to demonstrate radar-based planetary surface mapping concepts.

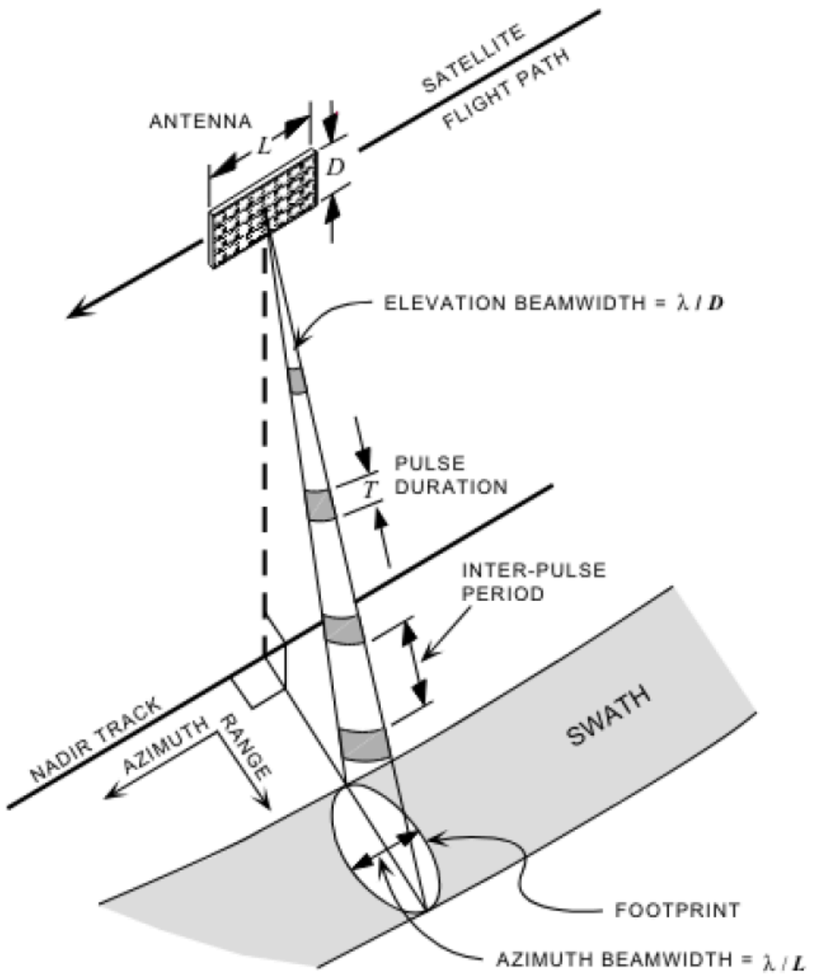
1 GHz – 8 GHz Double Ridged Horn Antenna



USRP B210


GPS ANT
10MHz
PPS
Ettus
Research
USRP
B210
RF B: TX/RX
RF B: RX2
RF A: RX2
RF A: TX/RX

SPECIFICATIONS:



Parameters	Values
Operating Frequency	1 to 6 GHz
Effective Aperture	$0.0209\,m^2$
Physical Aperture	$0.037\,m^2$
Bandwidth	500 MHz
Azimuth Beamwidth	43.5°
Elevation Beamwidth	75.5°
Antenna height from ground	2 m
Ground Swath Length	3.09 m
Ground Swath Width	1.6 m
Azimuth Resolution	0.076 cm
Range Resolution	0.3 m

APERTURE OF ANTENNA

Effective Aperture of Antenna

$$A_e = \frac{G\lambda^2}{4\pi}$$

$$\lambda = 0.15m \qquad [\ \lambda = \frac{c}{f}\ ]$$

$$G_{dBi} = 10.66 \qquad \text{for 2 GHz}$$

$$G = 11.65dB \qquad [\ G = 10^{\frac{G_{dBi}}{10}}\ ]$$

$$A_e = 0.0209m^2$$

Physical Aperture of Antenna

Width = 241 mm = 0.241 m

Height = 153 mm = 0.153 m

$$A_p = W \times H$$

$$A_p = 0.037m^2$$

Azimuth Beamwidth

$$\theta \approx \frac{70\lambda}{D}$$

D = 0.241 m

D => Largest dimensions of Antenna

$$\theta \approx 43.5^\circ$$

Elevation Beamwidth

$$D = 0.139m$$

$$\theta \approx 75.5^\circ$$

GROUND SWATH WIDTH

$$W = 2R\tan\frac{\theta}{2}$$

R = 2 m

W = 1.6 m

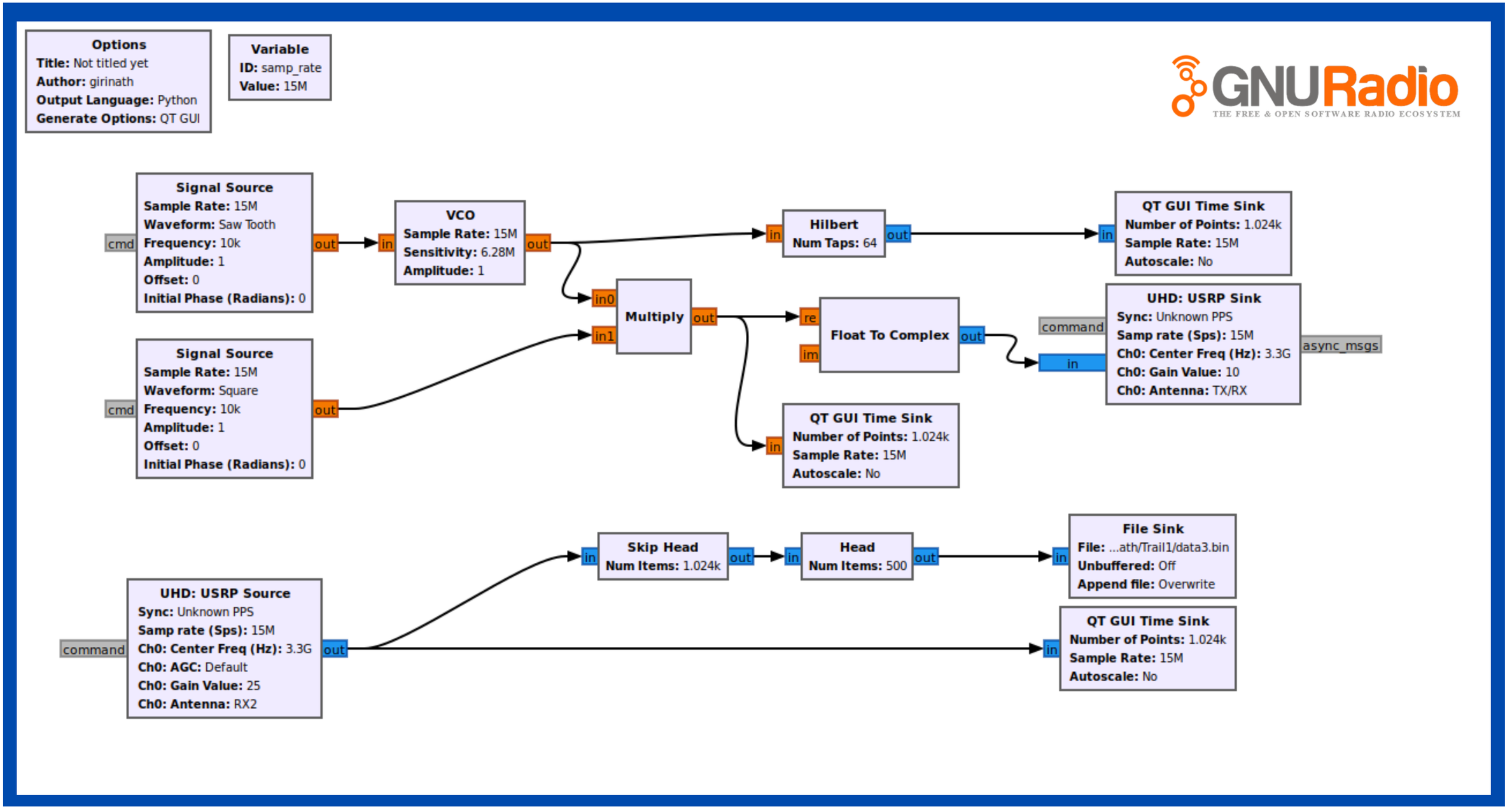
SAR AZIMUTH RESOLUTION

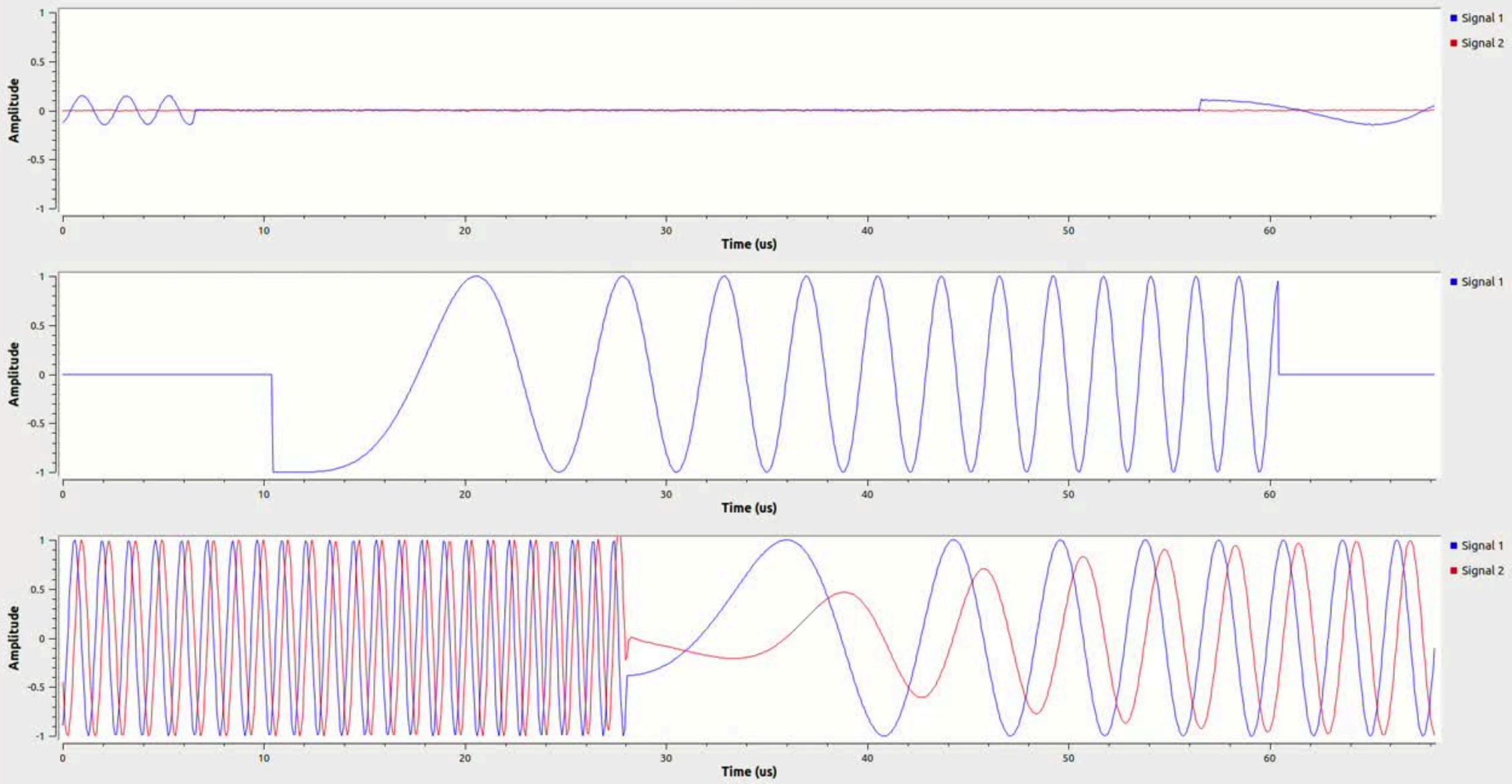
$$\rho_{az} = \frac{L}{2}$$

L = antenna length in the flight direction

L = 0.153 m

$$\rho_{az} = 0.0765m$$


Options
Title: Not titled yet
Author: girinath
Output Language: Python
Generate Options: QT GUI
Variable
ID: samp_rate
Value: 15M
GNURadio
THE FREE & OPEN SOFTWARE RADIO ECOSYSTEM
Signal Source
Sample Rate: 15M
Waveform: Saw Tooth
Frequency: 10k
Amplitude: 1
Offset: 0
Initial Phase (Radians): 0
cmd
out
VCO
Sample Rate: 15M
Sensitivity: 6.28M
Amplitude: 1
in
out
Hilbert
Num Taps: 64
QT GUI Time Sink
Number of Points: 1.024k
Sample Rate: 15M
Autoscale: No
Signal Source
Sample Rate: 15M
Waveform: Square
Frequency: 10k
Amplitude: 1
Offset: 0
Initial Phase (Radians): 0
Multiply
in0
in1
re
im
Float To Complex
command
UHD: USRP Sink
Sync: Unknown PPS
Samp rate (Sps): 15M
Ch0: Center Freq (Hz): 3.3G
Ch0: Gain Value: 10
Ch0: Antenna: TX/RX
async_msgs
QT GUI Time Sink
Number of Points: 1.024k
Sample Rate: 15M
Autoscale: No
Skip Head
Num Items: 1.024k
Head
Num Items: 500
File Sink
File: ...ath/Trail1/data3.bin
Unbuffered: Off
Append file: Overwrite
UHD: USRP Source
Sync: Unknown PPS
Samp rate (Sps): 15M
Ch0: Center Freq (Hz): 3.3G
Ch0: AGC: Default
Ch0: Gain Value: 25
Ch0: Antenna: RX2
QT GUI Time Sink
Number of Points: 1.024k
Sample Rate: 15M
Autoscale: No


Signal 1
Signal 2
Amplitude
Time (us)
1
0.5
0
-0.5
-1
0
10
20
30
40
50
60

RADAR Echo and IQ Sampling

- In Synthetic Aperture Radar, the radar transmits a waveform (usually an LFM chirp signal) toward the target.
- The reflected echo arrives after a time delay depending on the target distance.

- The received RF echo is converted to baseband IQ signals through demodulation:

$$s(t)=I(t)+jQ(t)$$

Component	Component
I	cosine part
Q	sine part
Complex signal	amplitude + phase

- The analog IQ signals are then sampled by an ADC, producing digital complex samples:

$$s[n]=I[n]+jQ[n]$$

- Each sample represents the radar echo at a specific time.

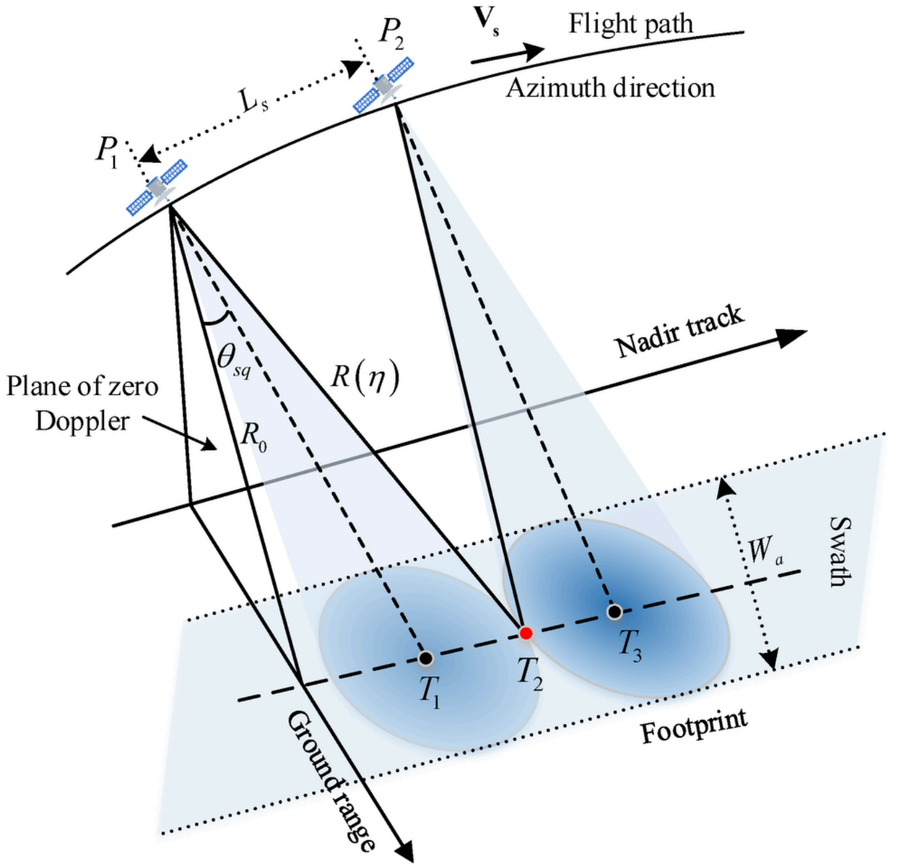
DATA Stitching in Synthetic Aperture RADAR

- Synthetic Aperture Radar forms a high-resolution image by combining echoes received from multiple radar positions.
- As the radar moves in the azimuth direction, it transmits pulses and records echoes from the ground.
- These echoes are arranged into a 2-D raw data matrix and later stitched together through SAR processing.

Pulse (Azimuth)	Range bin1	Range bin2	Range bin3	...
P1	echo	echo	echo	
P2	echo	echo	echo	
P3	echo	echo	echo	

RANGE CELL MIGRATION

- In Synthetic Aperture Radar, the radar platform moves along the azimuth direction while transmitting pulses.
- As the radar moves, the distance between the radar and a ground target changes continuously.
- Because of this changing distance, the echo from a single target does not remain in the same range bin.
- Instead, it appears in different range bins for different pulses.



RANGE CELL MIGRATION CORRECTION

Aligning and coherently combining radar echoes collected from multiple radar positions to synthesize a long antenna and focus targets into image pixels.

Before Range Cell Migration Correction

Pulse	Range bin1	Range bin2	Range bin3	Range bin4
P1			Target	
P2		Target		
P3	Target			
P4		Target		
P5			Target	

After Range Cell Migration Correction

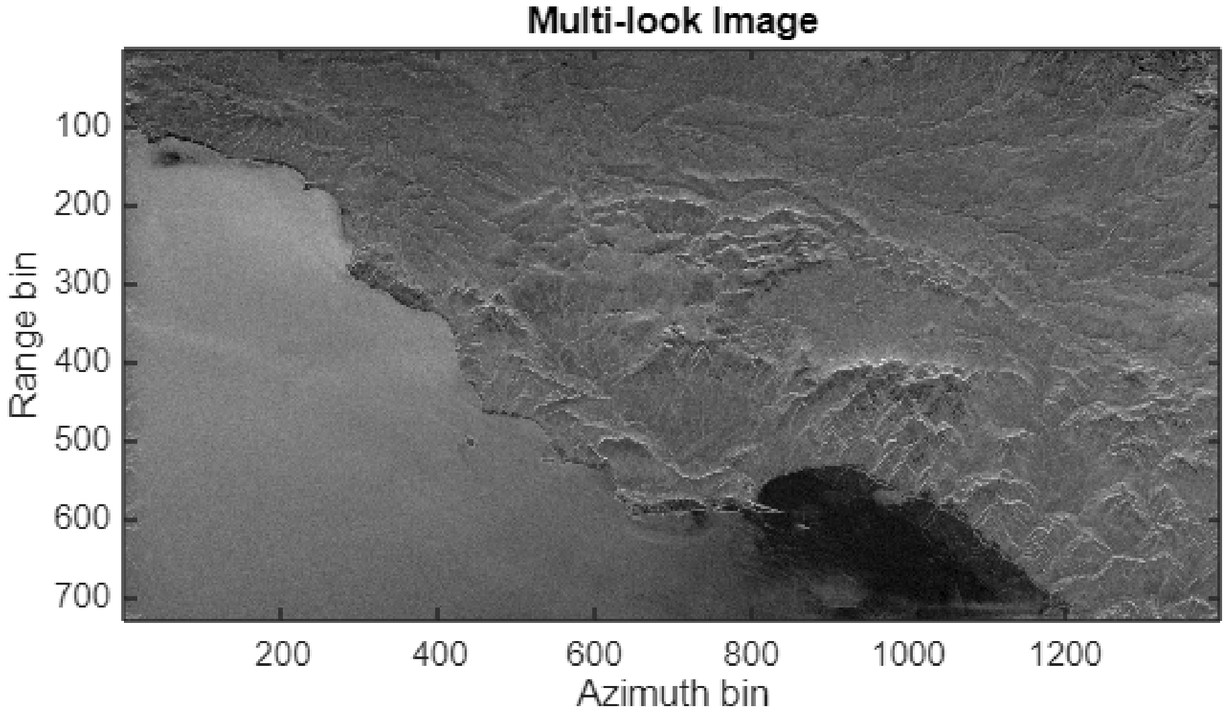
Pulse	Range bin1	Range bin2	Range bin3
P1		Target	
P2		Target	
P3		Target	
P4		Target	

The focused radar data is arranged into a 2-D image matrix where the horizontal axis represents azimuth and the vertical axis represents range, producing the final SAR image of the scene.

ERS SAR Raw Data Extraction and Image Formation

ERS dataset is NASA's provision of the European Space Agency's ERS satellites (ERS-1 and ERS-2) SAR data

Data Type: Level 1 GRD (Ground Range Detected)



Range migration image formation algorithm is used to focus the raw data.

Then Multi-look processing is applied to remove the multiplicative noise.

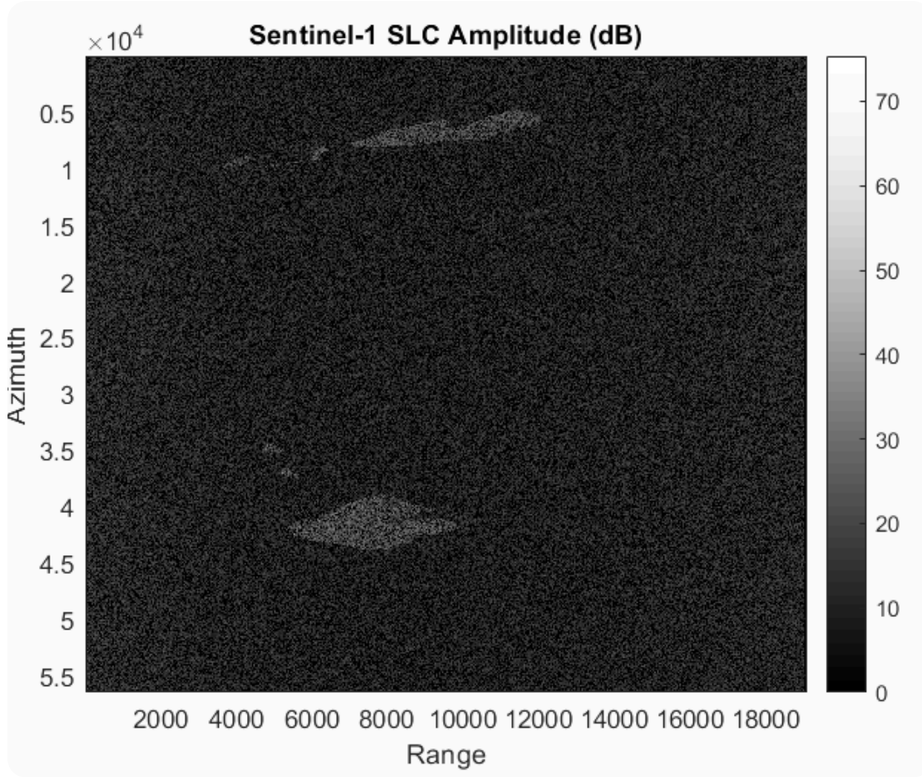
Reference : [1] Dataset: ERS-2, ESA 2011. Retrieved from ASF DAAC 10 November 2021.

Sentinel 1A C-band Synthetic Aperture Radar

SLC data from the European Space Agency (ESA) Sentinel 1A satellite
Data Type: Level 1 SLC (Single Look Complex) from the CEDA Archive



Processed with SNAP tool

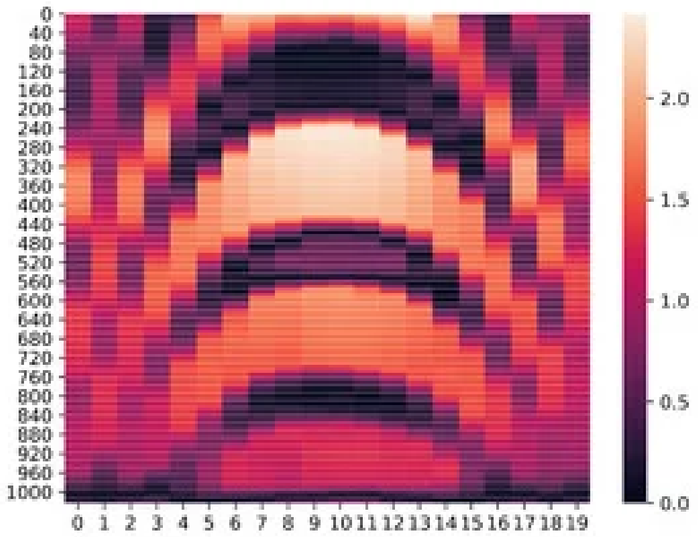


MATLAB figure

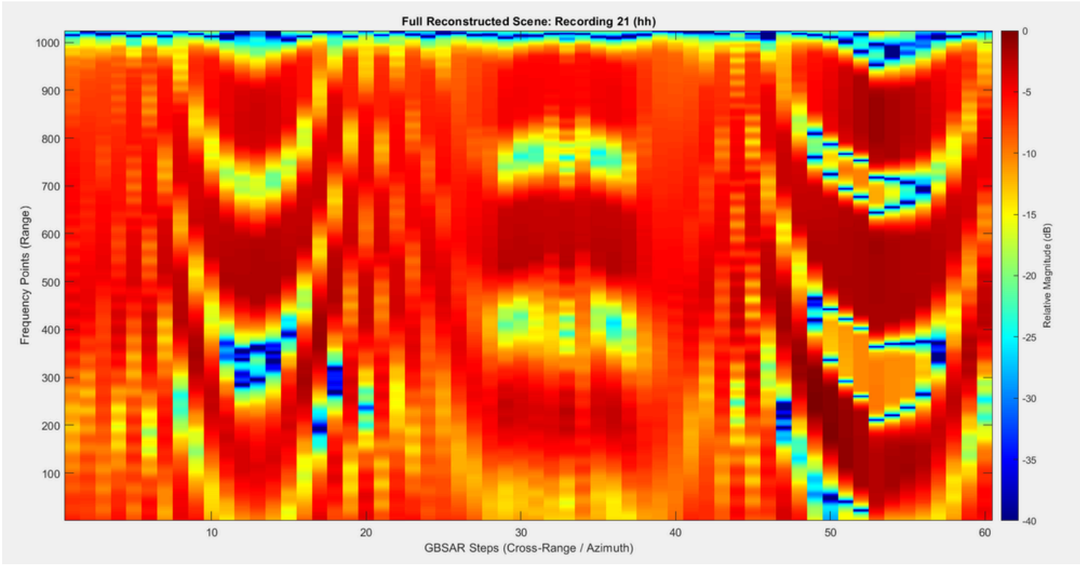
Reference : Copernicus; European Space Agency (2021): Sentinel 1A C-band Synthetic Aperture Radar (SAR): SM mode SLC Level 1 data, Instrument Processing Facility (IPF) v2. NERC EDS Centre for Environmental Data Analysis, date of citation. Turčinović, F.; Kačan, M.; Bojanjac, D.; Bosiljevac, M.; Šipuš, Z. Utilizing Polarization Diversity in GBSAR Data-Based Object Classification. Sensors 2024, 24, 2305. https://doi.org/10.3390/s24072305

GBSAR Data-Based Object Classification

- Three objects are set at an approximately 30 cm distance from the radar.
- The objects were of same size but made of different materials (metal, glass, plastic) and had different shapes (cylinder and cuboid).



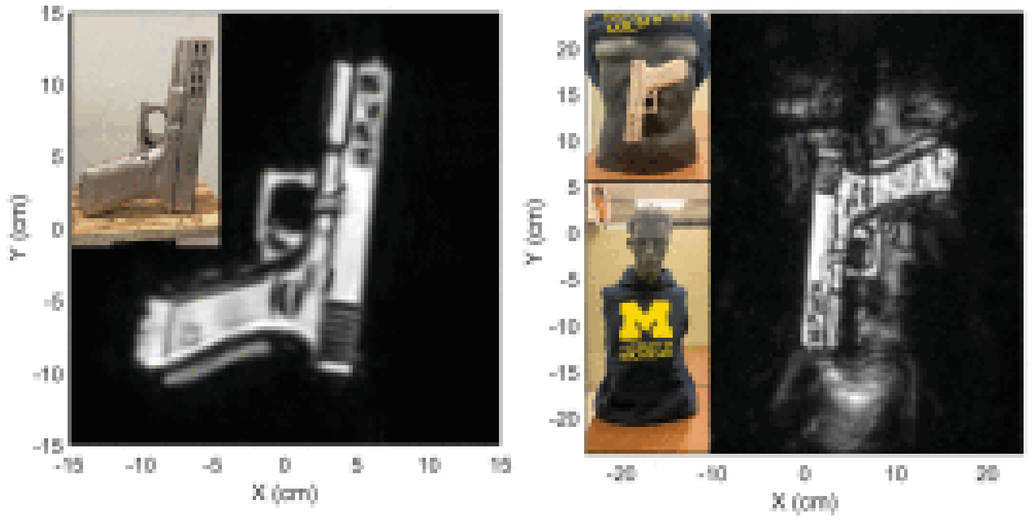
Reference image

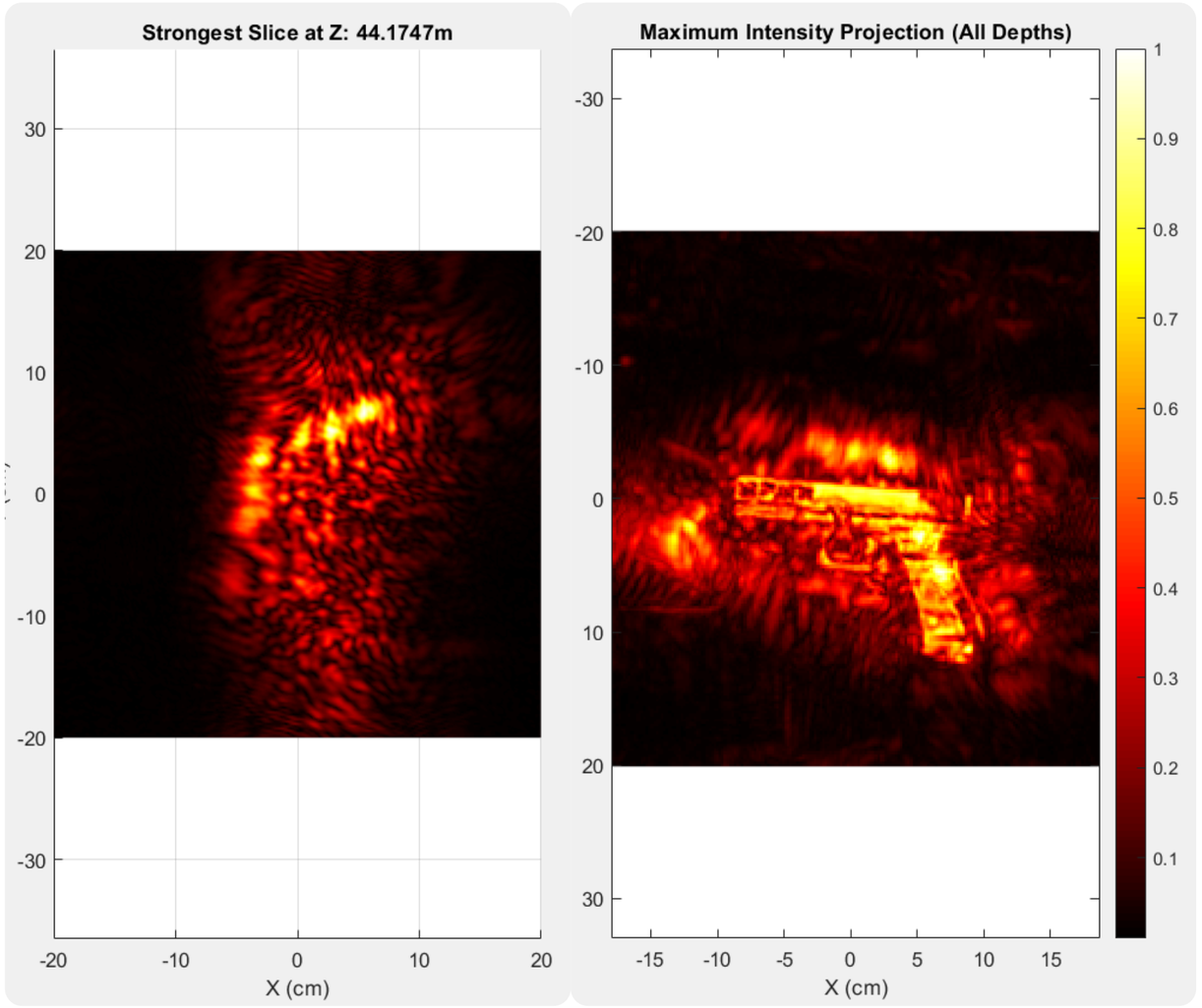


MATLAB figure

Reference : Turčinović, F.; Kačan, M.; Bojanjac, D.; Bosiljevac, M.; Šipuš, Z. Utilizing Polarization Diversity in GBSAR Data-Based Object Classification. Sensors 2024, 24, 2305. https://doi.org/10.3390/s24072305

FMCW-SAR Range Migration Algorithm Imaging





Reference : A. V. Muppala, A. Y. Nashashibi, E. Afshari and K. Sarabandi, "Fast-Fourier Time-Domain SAR Reconstruction for Millimeter-Wave FMCW 3-D Imaging," in IEEE Transactions on Microwave Theory and Techniques, vol. 72, no. 12, pp. 7028-7038, Dec. 2024, doi: 10.1109/TMTT.2024.3406938.

THANK YOU !